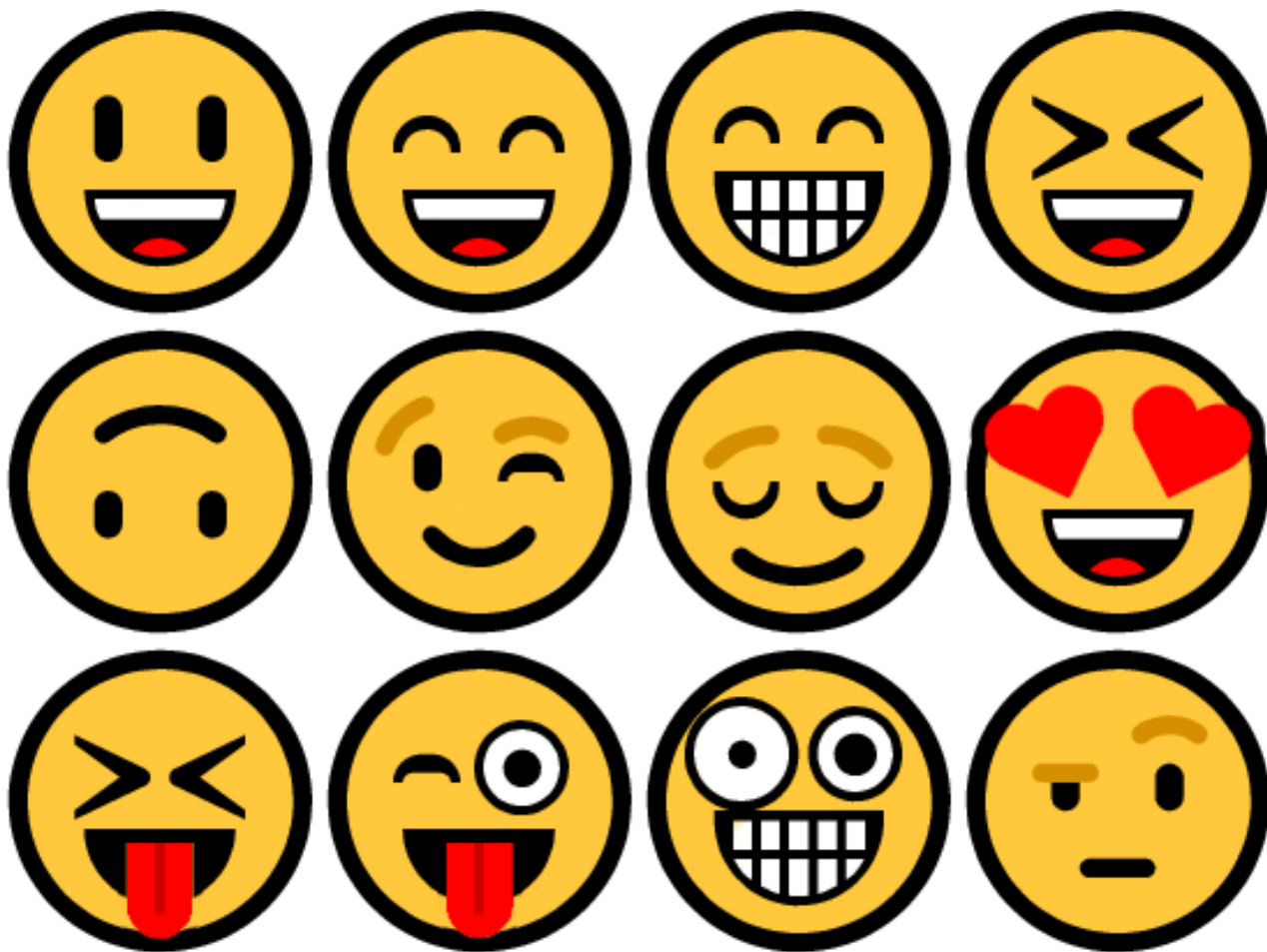




Emoji System Base

Documentation

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Overview

The **Emoji System Base** is an emoji library for **Unity**.

With it you can add a whole list of emojis to your games as well as a tool to make editing **SpriteAsset** and **TMP_SpriteAsset** easier.

Emoji to Unicode Converter

<https://lingoam.com/EmojiToUnicodeConverter>

Emoji	Unicode	Emoji	Unicode	Emoji	Unicode	Emoji	Unicode	Emoji	Unicode
	1f600		1f603		1f604		1f601		1f606
	1f605		1f602		1f923		263a		1f60a
	1f607		1f642		1f643		1f609		1f60c
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	1f614		1f61f		1f615		1f641		2639
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	1f625		1f613		1f917		1f914		1f92d
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	1f62f		1f626		1f627		1f62e		1f632
	1f634		1f924		1f62a		1f635		1f910
	1f974		1f922		1f92e		1f927		1f637
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	1F99B		1F98F		1F42A		1F42B		1F992
	1F998		1F403		1F402		1F404		1F40E
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Script Explanations

Emoji System Editor

This script creates a **custom editor tool inside Unity** for managing emoji sprite assets. Think of it as a **mini control panel** where you can:

- Edit emoji glyph settings (size, spacing, etc.)
- Switch between two systems (Legacy and TextMeshPro)
- Apply emoji assets to all text objects in a scene
- Cut a texture into multiple square images automatically

I'll walk you through it in a practical way: **what each variable represents, what each part does, and how you actually use it.**

1. Core Idea of the Tool

The editor window is divided into two main parts:

1. Asset Editor

Used to:

- Select an emoji sprite asset
- Modify how each emoji behaves (size, spacing, alignment)
- Save changes
- Apply changes to all text in the scene

2. Sprite Editor

Used to:

- Take a texture (like a sprite sheet)
- Cut it into equal square images
- Export them as individual files

2. Variables (What Each One Controls)

General UI State

- **scrollPosition** → Keeps track of scroll position in the window.
- **showAssetEditor** → Opens/closes the Asset Editor section.
- **showSpriteEditor** → Opens/closes the Sprite Editor section.

Asset Selection

- **emojiType** → Chooses between:
 - Legacy system (older Unity text system)
 - TextMeshPro (modern text system)
- **spriteAssetEdit** → The selected Legacy emoji asset.
- **TMP_spriteAssetEdit** → The selected TextMeshPro emoji asset.

 Only one of these is used depending on the selected type.

Glyph Metrics (VERY IMPORTANT)

These define how each emoji is rendered:

- **w (width)** → How wide the emoji is.
- **h (height)** → How tall the emoji is.
- **bx (bearing X)** → Horizontal offset from origin.
- **by (bearing Y)** → Vertical offset from baseline.
- **ad (advance)** → How much space to move after drawing the emoji.
- **scale** → Overall size multiplier.
- **atlasIndex** → Which texture atlas the emoji belongs to.

 These values affect alignment, spacing, and visual consistency.

Sprite Cutting

- **spriteEdit** → The texture you want to cut.
- **size** → Size (in pixels) of each square piece.

Internal Helpers

- **serializedObject** → Keeps the editor data synchronized with Unity.
- **textStyle / foldoutStyle** → Visual styles for the UI.

□ 3. Window Creation

What happens when you open the tool:

- A new window appears in Unity under:
- Tools → Emoji System Editor
- It tries to find an icon automatically from a specific folder.
- If found, the window gets a custom icon.

4. Interface Rendering (Main Loop)

This is where everything is drawn on screen.

It:

- Creates a scrollable layout
- Draws collapsible sections
- Shows input fields and buttons
- Enables/disables sections depending on conditions

5. Asset Editor Section

Step 1: Choose Asset Type

You pick:

- Legacy OR
- TextMeshPro

Step 2: Select Sprite Asset

Depending on type:

- You assign the asset you want to edit

Step 3: Edit Glyph Metrics

You manually input:

- Width / Height
- Bearings
- Advance
- Scale
- Atlas index

 These values will be applied to **ALL glyphs in the asset**

Step 4: Save Asset

When clicking **Save**:

1. It asks for confirmation
2. Applies your values to every glyph
3. Marks the asset as modified
4. Saves it permanently

 No undo after this.

Step 5: Apply to Scene (TMP Only)

If using TextMeshPro:

- It scans every object in the scene
- Finds all text components
- Replaces their emoji asset

 Useful for global updates.

6. Helper: Section Drawer

There's a reusable system that:

- Creates titled boxes
- Enables/disables interaction
- Keeps UI consistent

7. Glyph Editing Logic

What happens during save:

For each emoji glyph:

- Replace its metrics with your values
- Apply:
 - width/height
 - spacing
 - offsets
- Update:
 - scale
 - atlas index

👉 It overwrites everything uniformly.

8. Saving System

After editing:

- Marks asset as "dirty" (changed)
- Forces Unity to save it
- Refreshes the editor

9. Scene Application Logic

When applying to scene:

1. Finds all objects (even inactive ones)
2. Filters only those in current scene
3. Looks for text components
4. Assigns new emoji asset
5. Marks scene as modified

10. Sprite Cutting System

This is a **batch image processor inside Unity**.

How it works:

1. Loads the texture
 - From file OR
 - Creates a readable copy
2. Validates:
 - Texture exists
 - Size is valid
3. Calculates grid:
4. $\text{columns} = \text{width} / \text{size}$
5. $\text{rows} = \text{height} / \text{size}$
6. Loops through grid:
 - Extracts each square
 - Copies pixel data
 - Creates a new image
7. Saves each piece:
8. `square_row_col.png`
9. Cleans memory
10. Refreshes Unity

11. Texture Handling (Important Detail)

Textures in Unity are often **not readable**.

So the script tries:

Method 1:

Load directly from file (fast & accurate)

Method 2 (fallback):

Copy from GPU using a temporary buffer

👉 This ensures the tool works even with restricted textures.

12. How You Actually Use This Tool

✓ **Editing Emoji Assets**

1. Open the editor
2. Choose type (TMP recommended)
3. Assign sprite asset
4. Adjust metrics
5. Click **Save**

✓ **Updating Scene**

1. Select TMP asset
2. Click **Apply to Scene**

✓ **Cutting Sprite Sheets**

1. Assign texture
2. Set square size
3. Click **Apply to Folder**
4. Choose destination

⚠ **Important Observations**

- Glyph editing applies to **all emojis at once**
- No per-glyph editing (this is global)
- Saving is **permanent**
- Scene changes affect **every text object**

✂ **Final Summary**

This tool is basically:

A **multi-purpose emoji pipeline tool** inside Unity.

It combines:

- Asset editing
- Scene-wide updates
- Texture processing

All in one place, focused on **speed and batch operations** rather than fine manual editing.